

Linear Algebra

LAB Sheet - Lecture 1

Name: _____

ID/Batch: _____

Easy

Q. 1) Write the following system in matrix notation:

$$2x + y - z = 4$$

$$x - z = 1$$

$$3x + y + 2z = 6$$

Q. 2) Sketch

$$x + y = 3$$

$$x - y = 1$$

State the number of solutions.

Q. 3) Convert the following matrix equations back into systems of linear equations.

$$(a) \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$(b) \begin{bmatrix} 4 & 0 \\ -1 & 2 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 10 \end{bmatrix}$$

Q. 4) Draw the two pictures in two planes for the equations $x - 2y = 0$, $x + y = 6$.

Medium

Q. 5) Fill in the blanks:

(a) The intersection of two planes through (0,0,0) is probably a _____ but it could be a _____. It can't be the origin itself.

(b) The intersection of a plane through (0,0,0) with a line through (0,0,0) is probably a _____ but it could be a _____.

Q. 6) Starting with $x + 4y = 7$, find the equation for the parallel line through $x = 0$, $y = 0$. Find the equation of another line perpendicular to the first at $x = 3$, $y = 1$.

Q. 7) Sketch the following three lines:

$$2x + y = 5$$

$$x - y = 1$$

$$3x = 6$$

- (a) Does this system have a common solution?
- (b) What happens if all the right-hand sides become zero?
- (c) Is there any other non-zero choice of right-hand sides that allows all three lines to intersect at the same point? If yes, give one such example.

Q. 8) For the system

$$\begin{aligned} 2x + y &= 7 \\ x - 2y &= -1 \end{aligned}$$

draw the row picture and the column picture. Explain, in one or two sentences, how these two pictures represent the same mathematical problem from different perspectives.

Q. 9) Without solving the systems algebraically, try to answer whether each system has: exactly one solution, infinitely many solutions, or no solution.

- (a) $x + y = 3$; $2x + 2y = 6$
- (b) $x + y = 3$; $2x + 2y = 3$
- (c) $x + y = 3$; $x - y = 1$

Q. 10) A cafe sells only coffee and sandwiches. Two coffees and one sandwich cost ₹250. One coffee and three sandwiches cost ₹350.

- (a) Form a system of linear equations.
- (b) Solve using substitution or elimination method.
- (c) Write the corresponding matrix equation.
- (d) Explain the row picture of this system.

Q. 11) Consider

$$\begin{aligned} x - y &= 2 \\ 3x - 3y &= k \end{aligned}$$

Find the values of k for which the system has: (a) exactly one solution, (b) infinitely many solutions, and (c) no solution.

Explain your reasoning geometrically.

Hard

Q. 12) The following equations are certain to have the solution $x = y = 0$. Using row picture or column picture, determine for which values of a is there a whole line of solution?

$$\begin{aligned} ax + 2y &= 0 \\ 2x + ay &= 0 \end{aligned}$$

Q. 13) Explain why does the following system

$$\begin{aligned} u + v + w &= 2 \\ u + 2v + 3w &= 1 \\ v + 2w &= 0 \end{aligned}$$

have no solutions. What value should replace the last zero on the right side to allow the equations to have solution.

Q. 14) Consider the following linear system:

$$u \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + v \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + w \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Show that the three columns lie in the same plane by expressing the third column as a combination of the first two.

Q. 15) In the following equations, the third column (multiplying w) is the same as the right side. The column form of the equations immediately gives the solution for (u, v, w) ?

$$6u + 7v + 8w = 8$$

$$4u + 5v + 9w = 9$$

$$2u - 2v + 7w = 7$$

Challenge

Q. 16) Describe the intersection of the three planes $u + v + w + z = 6$ and $u + w + z = 4$ and $u + w = 2$ (all in four dimensional space.) Is it a line or a point or an empty set? What is the intersection if the fourth plane $u = -1$ is included? Find a fourth equation that leaves us with no solution.

Linear Algebra

LAB Sheet - Lecture 2

Name: _____

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Easy

Q.1) A startup manufactures three products: laptops, tablets and smart watches. Each column of

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 2 & 4 \end{bmatrix}$$

represents the amount of Steel, Plastic and Silicon required to produce one unit of each product. The production plan is

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

where the entries represent the number of laptops, tablets and smart watches to be produced. Compute the total amount of each raw material required given by $A\mathbf{x}$.

Q.2) A music recommendation system stores user preferences as $\mathbf{u}$ and song characteristics as A

$$\mathbf{u} = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 3 \end{bmatrix}$$

The row vector $\mathbf{s} = \mathbf{u}A$ gives the recommendation score assigned to each song. Compute $\mathbf{s}$.

Q.3) Tiny sound clips consisting of amplitudes of the Bach and MLK audios from the Lecture 0 have been stored in the matrix

$$S = \begin{bmatrix} 2 & 1 \\ -1 & 1 \\ 1 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} | & | \\ \text{Bach} & \text{MLK} \\ | & | \end{bmatrix}.$$

(a) Which recipe should produce both sounds at their original amplitudes?

$$\square \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \square \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \square \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

(b) Which recipe should reduce Bach to half amplitude while keeping MLK unchanged?

$$\square \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} \quad \square \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$$

(c) We represent several recipes at once by storing them as columns of another matrix

$$R = \begin{bmatrix} 1 & 1 & \frac{1}{2} \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} | & | & | \\ r_1 & r_2 & r_3 \\ | & | & | \end{bmatrix}$$

Without multiplying, interpret what happens with respect to each recipe column: sr_1, sr_2, sr_3 .

(d) Compute SR .

Medium

Q.4) A 5×5 matrix contains zeros everywhere except

$$a_{ii} = 1 \quad \text{and} \quad a_{i,5} = 2$$

for $i = 1, \dots, 4$. Let

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 5 \end{bmatrix}$$

Compute $A\mathbf{x}$ without constructing the full matrix. Explain your reasoning.

Q.5) Three stage lamps are pointed at a white wall. Each lamp's colour is stored as its red, green and blue intensities, and the three lamps become the three columns of

$$L = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{matrix} \leftarrow R \\ \leftarrow G \\ \leftarrow B \end{matrix} = \begin{bmatrix} | & | & | \\ \text{Red} & \text{Cyan} & \text{Yellow} \\ | & | & | \end{bmatrix}.$$

Each lamp has a dial running from 0 (off) to 1 (full). A **recipe** is a column x holding the three dial settings, and Lx is the colour that lands on the wall.

The three lamps — columns of L



Red lamp $(1, 0, 0)$



Cyan lamp $(0, 1, 1)$



Yellow lamp $(1, 1, 0)$

What the wall shows — columns of LR



white $(1, 1, 1)$



mid grey $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$



pale green $(\frac{1}{2}, 1, \frac{1}{2})$

(a) Which recipe turns the Red and Cyan lamps on at full power with Yellow switched off?

$$\square \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \square \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \square \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

What colour appears on the wall, and why is that colour not one of the three lamps?

(b) Which recipe dims the Red lamp to half brightness while leaving Cyan at full and Yellow off?

$$\square \begin{bmatrix} 1 \\ \frac{1}{2} \\ 0 \end{bmatrix} \quad \square \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}$$

- (c) Several recipes are stored side by side as the columns of

$$R = \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} | & | & | \\ r_1 & r_2 & r_3 \\ | & | & | \end{bmatrix}.$$

Without multiplying anything out, say what happens for each of Lr_1 , Lr_2 , Lr_3 . In particular: what is the relationship between r_1 and r_2 , and what does that force about the two colours they produce?

- (d) Look at the **second row** of L , namely $(0, 1, 1)$. What does this row describe? When you compute the second entry of Lr_1 , what quantity are you calculating? Answer in words, not symbols.
- (e) The dials can only add light — they can never remove it. Explain why, no matter how the three dials are set, the wall can never show pure blue $(0, 0, 1)$. (*Hint: look at the last two rows of L and ask which lamps contribute blue.*)
- Q.6)** A communication system receives signals from six antennas. Their signal strengths are stored in $\mathbf{x}$. The system does not process the six signals separately. Instead, it combines them into three aggregated channels using matrix A .

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \\ 10 \\ 12 \end{bmatrix}$$

where the entries correspond to Antennas 1, 2, ..., 6.

Thus, $A\mathbf{x}$ represents the three aggregated channel strengths.

- (a) Without carrying out the matrix multiplication, determine which two antennas contribute to the first aggregated channel? Which two contribute to the second and third channels?
- (b) An engineer says:

“If Antenna 4 becomes twice as strong, only one of the three aggregated channels will be affected.”

Do you agree? Which channel changes and why?

- (c) Suppose Antenna 2 suddenly stops transmitting. Which aggregated channel(s) will change?
- (d) Another engineer accidentally rearranges the antenna signals as

$$\mathbf{x}' = \begin{bmatrix} 2 \\ 6 \\ 4 \\ 8 \\ 10 \\ 12 \end{bmatrix}$$

Without multiplying, identify which aggregated channel(s) changes compared with $A\mathbf{x}$, and explain why.

- (e) The system designer wants to modify A so that the first aggregated channel combines Antennas 1 and 5 instead of Antennas 1 and 4. Which row of A must be changed, and what should that row become?

Hard

Q.7) Let A be the 3×3 matrix of **subjects**:

$$A = \begin{bmatrix} \emptyset & \emptyset & \text{Algebra} \\ \emptyset & \text{Matrix multiplication} & \emptyset \\ \text{Two planes} & \emptyset & \emptyset \end{bmatrix}$$

and B be the 3×3 matrix of **predicates**

$$\begin{bmatrix} \text{meet in a line} & \text{can be parallel} & \text{never meet at a point} \\ \text{rocks} & \text{combines the columns} & \text{takes rows times columns} \\ \text{is fun} & \text{explains everything} & \text{never lies} \end{bmatrix}$$

and C the 3×1 column of tails

$$\begin{bmatrix} \text{, and that is a fact} \\ \text{, as Strang shows} \\ \text{, believe it or not} \end{bmatrix}$$

Let us define the operation as below:

- $x \times y$: stick the words together with a space
- $x + y$: keep both as alternatives
- $\emptyset$: nothing at all; it wipes out whatever it touches, so $\emptyset \times y = \emptyset$

- (a) Which computations AB or BA is more meaningful? Compute and answer.
- (b) Compute $(AB)_{11}$ and $(BA)_{11}$. Use this to explain why $AB \neq BA$.
- (c) Compute $((AB)C)_{31}$.

Q.8) A social network contains 10 creators. Matrix A is a 10×10 influence matrix defined as:

- Every diagonal entry is 0.
- Every off-diagonal entry is 1.

Let

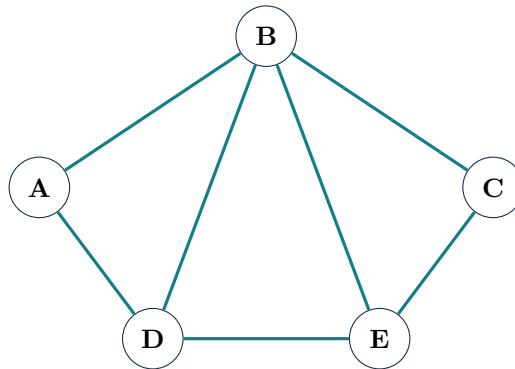
$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Without performing standard matrix multiplication:

- (a) Compute $A\mathbf{x}$.
- (b) Compute $A^2\mathbf{x}$.
- (c) Predict $A^3\mathbf{x}$.
- (d) Derive a formula for $A^k\mathbf{x}$.
- (e) Interpret your result in terms of repeated information propagation through a social-media network.

Challenge: Can a Matrix Count Routes?

Q.9) Consider the transportation network:



Let M be its adjacency matrix, where $m_{ij} = 1$ when one road directly connects i to j , and $m_{ij} = 0$ otherwise.

- Construct M using the vertex order A, B, C, D, E .
- Compute M^2 .
- Examine $(M^2)_{AE}$. List all walks from A to E using exactly two roads.
- Explain why each term $m_{Ak}m_{kE}$ tests whether k can be the middle stop.
- Complete:

$$(M^2)_{ij} = \text{number of } \underline{\hspace{10em}}.$$

- Predict what information is stored in M^3 .

Linear Algebra

LAB Sheet - Lecture 3

Name: _____

ID/Batch: _____

Easy

Q.1) Given two vectors:

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

(a) Compute $3v_1 + 2v_2$

(b) $3v_1 + 2v_2$ is a _____ while the set of all vectors of the form $av_1 + bv_2$ is called their _____.

Q.2) A linear combination produces _____ vector, whereas the span describes _____ vectors that can be produced.

Q.3) A drone has two basic movement commands:

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Without solving for the coefficients, decide which of the following targets could possibly be reached using equal amounts of the two commands:

$$\begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \quad \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

Then find the coefficients for the reachable target.

Q.4) A digital artist creates colours using two basis colours:

$$R = \begin{bmatrix} 255 \\ 0 \\ 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ 255 \\ 0 \end{bmatrix}.$$

(a) A student claims that yellow can be created by adding equal amounts of R and G. Do you agree? Write the corresponding linear combination and explain what the coefficients mean.

(b) Can a pure blue colour be created using only R and G ? (Think about this more!)

Q.5) A sailboat can move according to

$$v = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \quad \text{and} \quad w = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

Express

Three students make the following claims:

A: “The boat can reach every point in the plane.”

B: “The boat can reach every point whose x-coordinate is a multiple of 4 and y-coordinate is a multiple of 3.”

C: “The boat can reach only points in the first quadrant.”

Which claims are correct if arbitrary real coefficients are allowed? Explain.

Medium

Q.6) Consider the following three systems of two linear equations in two variables.

- $x + y = 5, \quad x - y = 1$
- $x + 2y = 3, \quad 2x + 4y = 6$
- $x + 2y = 3, \quad 2x + 4y = 7$

- (a) Draw both row picture and column picture of all the three equations.
- (b) For each of the linear system, can the two column vectors be scaled and added to produce the right-hand side vector $\mathbf{b}$.
- (c) Fill in the blanks.
 - (i) If $\mathbf{b}$ can be formed from the columns of $\mathbf{A}$ in exactly one way, the system has _____ solution.
 - (ii) If $\mathbf{b}$ can be formed from the columns of $\mathbf{A}$ in more than one way, the system has _____ solutions.
 - (iii) If $\mathbf{b}$ cannot be formed from the columns of $\mathbf{A}$, then $\mathbf{b}$ lies _____ the span of the columns of $\mathbf{A}$, and the system has _____ solution.
 - (iv) In the row picture, a solution corresponds to a point where the two _____.
 - (v) In the column picture, a solution corresponds to a way of mixing the columns of $\mathbf{A}$ to obtain _____.

Q.7) A rover can move according to

$$v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Without finding any coefficients:

- (a) Is their span a line or the entire plane?
- (b) What evidence from the vectors supports your answer?
- (c) Would replacing v_2

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

change your answer? Why?

Q.8) A song is described using three features: Energy, Danceability, Acousticness.

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}.$$

A recommendation engineer says:

“By adjusting the two coefficients, we can create any possible three-feature song profile.”

Do you agree?

You do not need to find coefficients. Use the structure of the two vectors to justify your answer.

Q.9) The growth of two species contributes

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}.$$

- (a) Before doing any calculations, what do you notice about v_1 and v_2 ? What does this immediately tell you about their span?
- (b) Can

$$\begin{bmatrix} 3 \\ 5 \\ 3 \end{bmatrix}$$

belong to the span? Justify your answer.

- (c) A student claims that because there are two vectors in $\mathbb{R}^3$, their span must be a plane. Is the student correct?

Q.10) Three toy drones occupy positions

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

- (a) Determine whether

$$\begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

can be expressed as a linear combination of these vectors in infinitely many ways. Explain why.

- (b) Why did infinitely many representations occur?
- (c) What relationship among the three movement vectors makes this possible?
- (d) If the third vector were removed, would infinitely many representations still be possible?

Hard

Q.11) Three sensors record vibration modes

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

An engineer says:

“We have three independent sensor modes in $\mathbb{R}^3$, so we can generate every possible vibration profile.”

Do you agree? You are not allowed to calculate a determinant or solve a system. Use the relationship among the three vectors to determine:

- (a) whether all three provide genuinely new directions,
- (b) whether their span is a line, plane or all of $\mathbb{R}^3$,

- (c) what happens if v_3 is removed.

Q.12) The movement commands available are

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

- (a) Express

$$\begin{bmatrix} 4 \\ 11 \\ 3 \end{bmatrix}$$

as a linear combination of the given vectors. Are the coefficients unique?

- (b) A student obtains one set of coefficients for the target. Another student obtains a different set. Both claim their answers are correct. Can both students be correct? Explain before calculating anything. What must be true about the three movement vectors for this to happen? Now verify by finding two different coefficient sets.

Q.13) A chemist can combine three reaction pathways represented by

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}.$$

Another chemist replaces v_3 with

$$\begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}$$

Without solving any system:

- (a) Which set spans a plane?
 (b) Which set can span all of $\mathbb{R}^3$.
 (c) What changed structurally between the two sets?

Challenge

Q.14) The Infinite Monkey Drone Controller

Recall the Infinite Monkey Theorem.

A monkey is given control of a drone whose only movement commands are

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

The monkey can choose **any real coefficients**, positive or negative, and combine the commands indefinitely.

- (a) Describe the set of all locations the drone can reach.
 (b) The monkey accidentally deletes one of the three movement commands. You are told only that the drone can still reach exactly the same set of locations as before. Which command could the monkey have deleted? Justify your answer without solving a system.

- (c) Suppose instead the monkey deletes v_1 . What new information would you need to determine whether the reachable region changed?
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Reflection:

Which application in this sheet surprised you the most? How does the idea of a linear combination appear in different fields such as robotics, biology, graphics, music recommendation systems, and chemistry?

Linear Algebra

LAB Sheet - Lecture 4

Name: _____

ID/Batch: _____

Easy

Q.1) A robot moving on a factory floor receives the following navigation commands:

$$\begin{aligned}x + y + z &= 9 \\2x + y + 3z &= 16 \\x + 2y + z &= 11\end{aligned}$$

Use Gaussian Elimination to determine the robot's final coordinates (x, y, z) .

Q.2) Apply Gaussian Elimination to obtain the Row Echelon Form of

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 2 \end{bmatrix}$$

How many pivots does the matrix contain?

Q.3) drone records measurements from four sensors:

$$B = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 1 \\ 1 & 2 & 1 & 1 \end{bmatrix}$$

Reduce B to Row Echelon Form and determine its rank.

Medium

Q.4) A small online store tracks revenue from three products:

$$\begin{aligned}x + y + z &= 120 \\2x + 3y + z &= 230 \\x + 2y + 4z &= 270\end{aligned}$$

where x, y, z represent the number of units sold. Use Gaussian Elimination to determine x, y, z .

Q.5) A quadcopter has four motors producing thrusts x_1, x_2, x_3, x_4 .

The following equilibrium equations are obtained:

$$\begin{aligned}x_1 + x_2 + x_3 + x_4 &= 40 \\x_1 - x_2 + x_3 - x_4 &= 0 \\x_1 + x_2 - x_3 - x_4 &= 4 \\x_1 - x_2 - x_3 + x_4 &= 8\end{aligned}$$

Solve using Gaussian Elimination.

Q.6) Find the rank of the rectangular matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 6 & 8 & 10 \\ 1 & 1 & 2 & 3 & 4 \\ 3 & 5 & 8 & 11 & 14 \end{bmatrix}$$

Q.7) A chemistry laboratory records concentrations of four compounds in four experiments:

$$\begin{aligned} x + y + z + w &= 18 \\ 2x + 3y + z + 2w &= 30 \\ x + 2y + 3z + w &= 25 \\ 3x + 4y + 4z + 3w &= 48 \end{aligned}$$

Use Gaussian Elimination to solve for all concentrations.

Q.8) Reduce the following 3×5 matrix to Row Echelon Form and determine its rank.

$$A = \begin{bmatrix} 1 & 1 & 2 & 1 & 0 \\ 2 & 3 & 5 & 3 & 1 \\ 1 & 2 & 3 & 2 & 1 \end{bmatrix}$$

Identify the pivot columns.

Hard

Q.9) A delivery company operates between five warehouses. The shipment variables x_1, x_2, x_3, x_4, x_5 satisfy

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 + x_5 &= 30 \\ 2x_1 + x_2 + 3x_3 + x_4 + x_5 &= 50 \\ x_1 + 2x_2 + x_3 + 2x_4 + x_5 &= 40 \\ 3x_1 + 2x_2 + 4x_3 + 3x_4 + 2x_5 &= 70 \\ x_1 + x_2 + 2x_3 + x_4 + 3x_5 &= 45 \end{aligned}$$

Solve using Gaussian Elimination.

Q.10) A machine-learning dataset contains measurements from six features across four observations:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 6 & 8 & 10 & 12 \\ 1 & 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 8 & 11 & 14 & 17 \end{bmatrix}$$

Find the rank.

Q.11) A robotic arm is controlled by four joint parameters.

$$\begin{aligned}x + y + z + w &= 12 \\2x + 3y + z + 2w &= 22 \\x + 2y + 3z + w &= 20 \\3x + 4y + 4z + 3w &= 34\end{aligned}$$

Solve using Gaussian Elimination and verify the solution by substitution.

Q.12) For the matrix

$$A = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 2 & 6 \\ 1 & 3 & 2 & 4 \\ 3 & 7 & 4 & 10 \end{bmatrix}$$

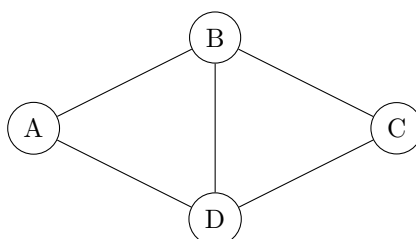
- Find the Row Echelon Form.
- Determine the rank.

Challenge Problem: Graph Theory and Matrices

Recall the graph theoretic application of matrices that we learnt in the second lecture. We shall now study one more such application to graph theory that is **Incidence matrix**

Example: Incidence Matrix

We had considered an example graph as follows:



Now label the edges as

$$\begin{aligned}e_1 &: A - B & e_2 &: A - D \\e_3 &: B - C & e_4 &: B - D \\e_5 &: C - D\end{aligned}$$

The incidence matrix records which vertices participate in each edge.

$$b_{ij} = \begin{cases} 1, & \text{if vertex } i \text{ lies on edge } e_j \\ 0, & \text{otherwise} \end{cases}$$

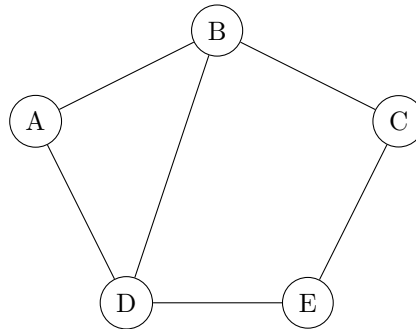
The incidence matrix becomes

$$B = \begin{array}{c|ccccc} & e_1 & e_2 & e_3 & e_4 & e_5 \\ \hline A & 1 & 1 & 0 & 0 & 0 \\ B & 1 & 0 & 1 & 1 & 0 \\ C & 0 & 0 & 1 & 0 & 1 \\ D & 0 & 1 & 0 & 1 & 1 \end{array}$$

Observe:

- Rows correspond to vertices.
- Columns correspond to edges.
- Each column contains exactly two ones because every edge connects two vertices.

Part C Consider the following transportation network:



1. Label the edges and construct the incidence matrix.
2. Determine the rank of the incidence matrix.
3. What does the rank tell you about the structure of the network?

Linear Algebra

LAB Sheet - Lecture 5

Name: _____

ID/Batch: _____

Easy

Q.1) Fill in the blanks:

Row Operations	Elementary Matrix E	Inverse Elementary E^{-1}
$R_2 \rightarrow R_2 - R_1$	_____	_____
_____	$\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$	_____
_____	_____	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$
$R_3 \rightarrow R_3 + 3R_2$	_____	_____
_____	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	_____

Q.2) An elementary matrix E and a matrix A are given. Identify the row operation corresponding to E and verify that the product EA results from applying the row operation to A .

(a) $E = \begin{bmatrix} -6 & 0 \\ 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$

(b) $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

Q.3) Consider a matrix A which After a certain elementary row operation is changed to matrix B .

$$A = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 1 & 4 \\ -3 & 2 & -10 \\ 1 & 2 & 3 \end{bmatrix}$$

Find the corresponding elementary matrix E and verify $EA = B$.

Q.4) Find a matrix E which when multiplied with $A = \begin{bmatrix} 6 & 2 & 1 \\ 3 & 5 & 2 \\ 2 & 1 & 4 \end{bmatrix}$ reduces it to row echelon form.

Medium

Q.5) State whether the following statements are True/False:

(a) The product of two 3×3 lower triangular matrices is also a lower triangular matrix.

- (b) If $A = LU$, then L and U are always unique.
- (c) If a 3×3 matrix A has an LU decomposition with three nonzero pivots in U , then $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbb{R}^3$.

Q.6) Find the LU decomposition of following matrices and verify $A = LU$

$$(a) A = \begin{bmatrix} 4 & 5 & 3 \\ 6 & 8 & 7 \\ 2 & 1 & 1 \end{bmatrix}, \quad (b) B = \begin{bmatrix} 2 & 1 & 1 & 3 \\ 4 & 5 & 3 & 7 \\ 6 & 8 & 8 & 15 \\ 2 & 7 & 9 & 18 \end{bmatrix}$$

Q.7) Compute L and U for the symmetric matrix

$$A = \begin{bmatrix} a & a & a \\ a & b & b \\ a & b & c \end{bmatrix}$$

Find three conditions on a, b, c to get $A = LU$ with three pivots.

Q.8) A robotics engineer has forgotten the original matrix A . After three row operations, the matrix becomes

$$U = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

The elementary matrices used were:

$$E_1 : R_2 \rightarrow R_2 - R_1, \quad E_2 : R_3 \rightarrow R_3 - 2R_1, \quad E_3 : R_3 \rightarrow R_3 + R_2$$

Without performing Gaussian elimination, recover the original matrix A .

Q.9) The same three stage lamps are pointed at the white wall. Each lamp's colour is stored as its red, green and blue intensities, and the three lamps are the three columns of

$$M = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{matrix} \leftarrow R \\ \leftarrow G \\ \leftarrow B \end{matrix} = \begin{bmatrix} | & | & | \\ \text{Red} & \text{Cyan} & \text{Yellow} \\ | & | & | \end{bmatrix}.$$



Red lamp (1, 0, 0)



Cyan lamp (0, 1, 1)



Yellow lamp (1, 1, 0)

A recipe is a column $\mathbf{x}$ of the three dial settings, and $M\mathbf{x}$ is the colour on the wall.

Rehearsals run every night and the director calls out a different colour each time, but the three lamps never change. So instead of repeating Gaussian elimination on M every night, do it *once* and store the work as $M = LU$.

Tonight the director asks for **amber**, $\mathbf{b} = (4, 3, 1)$:



Amber (4, 3, 1)

- (a) Apply Gaussian elimination on M , write LU -decomposition of M .
- (b) The system $M\mathbf{x} = \mathbf{b}$ is now $L(U\mathbf{x}) = \mathbf{b}$. Put $\mathbf{c} = U\mathbf{x}$ and solve $L\mathbf{c} = \mathbf{b}$ by forward substitution (top row first).
- (c) Solve $U\mathbf{x} = \mathbf{c}$ by back substitution (bottom row first) to get the recipe.

- (d) Tomorrow the director wants **sage green**, $\mathbf{b}' = (2, 3, 2)$:



sage green $(2, 3, 2)$

Find its recipe **without** performing a single new row operation on M .

- (e) What is actually recorded in L , and why does having it let you skip the elimination in part (d)? Answer in words.
- (f) Last week the director asked for pure blue, $(0, 0, 1)$. Feed it through the same L and U . You still get exactly one recipe. Write it down, and explain what goes wrong.

- Q.10)** If A and B have nonzeros in the positions marked by x , which zeros are still zero in their factors L and U ?

$$A = \begin{bmatrix} x & x & x & x \\ x & x & x & 0 \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix} \quad B = \begin{bmatrix} x & x & x & 0 \\ x & x & 0 & x \\ x & 0 & x & x \\ 0 & x & x & x \end{bmatrix}$$

Hard

- Q.11)** A logistics company operates four warehouses W_1, W_2, W_3 , and W_4 to distribute essential supplies to multiple cities during a festival season. The relationship between the truckloads dispatched from each warehouse and the transportation requirements is modeled by $A\mathbf{x} = \mathbf{b}$, where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

represents the number of truckloads dispatched from warehouses W_1, W_2, W_3, W_4 respectively. A is the transportation constraint matrix.

The company is considering two delivery plans:

- Plan 1 ($\mathbf{b}_1$): Prioritizes faster delivery to urban areas.
- Plan 2 ($\mathbf{b}_2$): Prioritizes balanced delivery across all regions.

The transportation constraint matrix is

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 4 & 5 & 3 & 1 \\ 2 & 3 & 4 & 2 \\ 6 & 7 & 8 & 5 \end{bmatrix}$$

The demand vectors for the two plans are

$$\mathbf{b}_1 = \begin{bmatrix} 4 \\ 13 \\ 11 \\ 26 \end{bmatrix} \quad \text{and} \quad \mathbf{b}_2 = \begin{bmatrix} 6 \\ 17 \\ 13 \\ 32 \end{bmatrix}$$

The total fuel consumption for delivery is given as

$$\text{Fuel Consumption} = 8x_1 + 6x_2 + 9x_3 + 5x_4$$

Which delivery plan is more fuel-efficient?

- Q.12)** A city's water authority operates three pumping stations that supply water to three major zones of the city. Due to differences in pipe lengths, elevation, and network connectivity, each pumping station contributes a different amount of water to each zone for every unit of pumping effort. The relationship between the pumping efforts and the amount of water delivered to the zones is modeled by $A\mathbf{x} = \mathbf{b}$ where,

$$A = \begin{bmatrix} 4 & 2 & 1 \\ 2 & 5 & 2 \\ 1 & 2 & 4 \end{bmatrix}$$

$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ represent the pumping effort (in hundreds of cubic metres per hour) at Pumping Stations 1, 2 and 3, respectively, whereas, Each entry of $A\mathbf{x}$ gives the amount of water delivered to one of the three city zones.

The city experiences different water demands throughout the day:

$$\mathbf{b}_1 = \begin{bmatrix} 7 \\ 8 \\ 13 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 10 \\ 16 \\ 13 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 7 \\ 9 \\ 7 \end{bmatrix}$$

corresponding to morning, afternoon, and evening demand, respectively. Using the LU decomposition, determine the pumping effort required at each station to satisfy the daily demands.

Challenge

- Q.13)** A chemical manufacturing company produces four specialty cleaning solutions by mixing four base chemicals: C_1, C_2, C_3 , and C_4 . Each litre of a base chemical contributes different amounts of four important properties of the final product: Cleaning Strength, Viscosity, pH Balance, Surface Protection. Through laboratory testing, engineers have determined that the relationship between the quantities of the base chemicals and the resulting properties is $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{bmatrix} 2 & 1 & 3 & 2 \\ 4 & 6 & 5 & 9 \\ -2 & 11 & -3 & 15 \\ 8 & -4 & 17 & 6 \end{bmatrix}$$

$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4$ denote the number of litres of chemicals C_1, C_2, C_3 , and C_4 , respectively. The matrix A describes how each litre of a chemical contributes to the four measurable properties. The vector $\mathbf{b}$ specifies the desired property values for a particular product.

The company wants to manufacture three different products: Glass Cleaner, Floor Cleaner, Industrial Degreaser.

$$\mathbf{b}_1 = \begin{bmatrix} 8 \\ 24 \\ 21 \\ 27 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 10 \\ 28 \\ 19 \\ 35 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 9 \\ 30 \\ 32 \\ 23 \end{bmatrix}$$

- (a) Complete the following production sheet.

Base chemical	Glass Cleaner	Floor Cleaner	Industrial Degreaser
C_1			
C_2			
C_3			
C_4			

- (b) Suppose the storage tank for Chemical C_2 can dispense at most 1 litre per batch. Based on your three computed recipes, which products can be manufactured immediately?

- (c) The company currently has enough chemicals to prepare only one product at a time. The total chemical used for a single product is calculated as:

$$\text{Total Chemical Used} = x_1 + x_2 + x_3 + x_4$$

Determine which product should be manufactured that will have least total chemical consumption.

- (d) The company has received orders for: 20 batches of Glass Cleaner, 15 batches of Floor Cleaner, 10 batches of Industrial Degreaser. Using the recipes you computed, determine the total quantity of each base chemical required. The warehouse currently contains:

Chemical	C_1	C_2	C_3	C_4
Available (in L)	60	40	50	55

Determine whether the current inventory is sufficient. If not, calculate the additional quantity of each chemical that must be ordered.

Linear Algebra

LAB Sheet - Lecture 6

Name: _____

ID/Batch: _____

Easy

Q.1) Show that the vectors $\mathbf{v}_1 = (1, 2, 3, 4)$, $\mathbf{v}_2 = (0, 1, 0, -1)$, and $\mathbf{v}_3 = (1, 3, 3, 3)$ form a linearly dependent set in $\mathbb{R}^4$. Express $\mathbf{v}_3$ as the linear combination of the other two.

Q.2) State whether the following statements are True/False:

- (a) A set containing a single vector is linearly independent.
- (b) The columns of any 4×5 matrix are linearly dependent.
- (c) No linearly independent set contains the zero vector.
- (d) Two vectors are linearly dependent if and only if they lie on a line through the origin.

Q.3) Determine whether $\mathbf{b}$ is in the column space of A using the rank condition, and if so, express $\mathbf{b}$ as a linear combination of the column vectors of A .

(a) $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$

(b) $A = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$

Q.4) Determine whether the following vectors are linearly independent or linearly dependent. Further, geometrically explain the span of these vectors.

- (a) $\mathbf{v}_1 = (2, -2, 0)$, $\mathbf{v}_2 = (6, 1, 4)$, $\mathbf{v}_3 = (2, 0, -4)$ in $\mathbb{R}^3$
- (b) $\mathbf{v}_1 = (-1, 2, 3)$, $\mathbf{v}_2 = (2, -4, -6)$, $\mathbf{v}_3 = (-3, 6, 0)$ in $\mathbb{R}^3$
- (c) $\mathbf{v}_1 = (4, 6, 8)$, $\mathbf{v}_2 = (2, 3, 4)$, $\mathbf{v}_3 = (-2, -3, -4)$ in $\mathbb{R}^3$
- (d) $\mathbf{v}_1 = (3, 8, 7, -3)$, $\mathbf{v}_2 = (1, 5, 3, -1)$, $\mathbf{v}_3 = (2, -1, 2, 6)$, $\mathbf{v}_4 = (4, 2, 6, 4)$ in $\mathbb{R}^4$

Medium

Q.5) Fill in the blanks:

- (a) If A is any 8 by 8 invertible matrix, then its column space is _____.
- (b) If A is an $m \times n$ matrix, then the columns of A are linearly independent if and only if A has _____ pivot columns.
- (c) If the 9 by 12 system $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$, then $C(A) =$ _____.
- (d) $C(AB)$ must _____ (contain $\supseteq$ / be contained in $\subseteq$ / equal $=$) the column space of $(A$ or $B)$ for all 4×4 matrices A and B .
- (e) If A is a 4×3 matrix and $A\mathbf{x} = \mathbf{b}$ is not solvable for some $\mathbf{b}$ and the solutions are not unique when they exist, possible values for the rank of A are _____ (list all possibilities).

Q.6) For which real values of λ do the following vectors form a linearly dependent set in $\mathbb{R}^3$?

$$\mathbf{v}_1 = (\lambda, -\frac{1}{2}, -\frac{1}{2}), \quad \mathbf{v}_2 = (-\frac{1}{2}, \lambda, -\frac{1}{2}), \quad \mathbf{v}_3 = (-\frac{1}{2}, -\frac{1}{2}, \lambda)$$

Q.7) State whether the following statements are True or False. Give proper **counter example** if the statement is false.

- (a) If $C(A)$ contains only the zero vector, then A is the zero matrix.
- (b) The column space of $2A$ equals the column space of A .
- (c) The column space of $A - I$ equals the column space of A .
- (d) If the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent, then $\{k\mathbf{v}_1, k\mathbf{v}_2, k\mathbf{v}_3\}$ is also linearly independent for every nonzero scalar k .
- (e) If $\mathbf{v}_1, \dots, \mathbf{v}_4$ are in $\mathbb{R}^4$ and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent, then $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is also linearly dependent.
- (f) If $\mathbf{v}_1, \dots, \mathbf{v}_4$ are in $\mathbb{R}^4$ and $\mathbf{v}_3$ is not a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$, then $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is linearly independent.

Q.8) Construct a 3 by 3 matrix whose column space contains $(1, 1, 0)$ and $(1, 0, 1)$ but not $(1, 1, 1)$. Further, also construct a 3 by 3 matrix whose column space is only a line.

Q.9) How many pivot columns must a 7×5 matrix have if its columns are linearly independent? Why? How many pivot columns must a 5×7 matrix have if its columns span $\mathbb{R}^5$? Why?

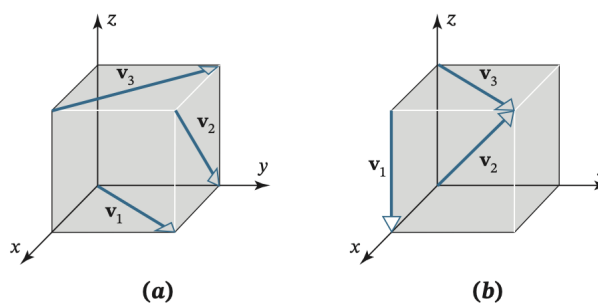
Q.10) For what values of h is $\mathbf{v}_3$ in $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$, and for what values of h is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly dependent? Justify each answer.

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -6 \\ 7 \\ -3 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 8 \\ h \\ 4 \end{bmatrix}$$

Hard

Q.11) The columns of AB are combinations of the columns of A . This means: The column space of AB is contained in (possibly equal to) the column space of A . Give an example where the column spaces of A and AB are not equal.

Q.12) Determine whether the vectors $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ are linearly independent or linearly dependent.



Q.13) For which right-hand sides (find a condition on b_1, b_2, b_3) are these systems solvable?

$$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Q.14) Your team is building a college robot with four control mechanisms. Their movement effects along three axes are given by the columns of:

$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 2 & 3 \end{bmatrix}.$$

To save money and simplify the build, you want to eliminate any controls that don't add new movement capabilities.

- How will you determine the redundant control mechanisms?
- Which control mechanisms are redundant?
- What is the smallest number of control mechanisms the team needs to retain to preserve all possible movement patterns that the full robot could produce?
- If the team keeps control mechanisms 1 and 2, can it still produce every movement pattern that it could produce using all four controls?

Challenge

Q.14) In previous labs, the theatre had three lamps, and every colour had exactly one recipe. The theatre has now bought a fourth lamp. Written as RGB columns, the four lamps are

$$L = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{matrix} \leftarrow R \\ \leftarrow G \\ \leftarrow B \end{matrix} = \begin{bmatrix} | & | & | & | \\ \text{Red} & \text{Cyan} & \text{Yellow} & \text{White} \\ | & | & | & | \end{bmatrix}$$



Red (1, 0, 0)



Cyan (0, 1, 1)



Yellow (1, 1, 0)



White (1, 1, 1)

A **recipe** $\mathbf{x}$ gives the brightness of each lamp, and $L\mathbf{x}$ is the colour on the wall. The set of all colours the rig can make is $\text{Col}(L)$.

- Apply Gaussian elimination to L . How many pivots are there, and what is the rank?
- Are the four lamp colours linearly independent? Without calculating, give a one-line reason using only the shape of L .
- Find four numbers, not all zero, so that multiplying the four lamp colours by them and adding gives (0, 0, 0).
- A student says: “*White is the extra lamp, so return White.*”
 - Using (c), show that the theatre could return **Red** instead and still make every colour it makes now.
 - Using linear independence, show that returning **Yellow** is not safe: the three lamps left over can only make colours whose green and blue amounts are equal.
- Last lecture the three-lamp rig made amber 
 $b = (4, 3, 1)$ from the recipe (2, 1, 2). On the new rig this recipe is $x = (2, 1, 2, 0)$.
 - Check that $L\mathbf{x} = b$.

- (ii) Part (c) says Red and Cyan together can be traded for White. Use this trade to write down a *different* recipe that also gives amber, and verify it.
- (iii) If brightness may be any real number, how many recipes give amber?
- (iv) In one sentence: what changed from last lecture to destroy the one-recipe-per-colour rule?
- (f) *True or false? Give a reason for each.*
 - (i) Buying the fourth lamp increased the number of colours the theatre can make.
 - (ii) Any four vectors in $\mathbb{R}^3$ are linearly dependent.
 - (iii) Any three of these four lamps are linearly independent.

Linear Algebra

LAB Sheet - Lecture 7

Name: _____

ID/Batch: _____

Easy

Q.1) State True/False. Justify your answer.

- (a) The null space of A is the solution set of the equation $A\mathbf{x} = 0$.
- (b) A square matrix has no free variables.
- (c) An invertible matrix has no free variables.
- (d) The null space of a $m \times n$ matrix is a subspace of $\mathbb{R}^m$.
- (e) An $m \times n$ matrix has no more than n pivot variables.
- (f) An $m \times n$ matrix has no more than m pivot variables.
- (g) The set of all solutions of a system of m homogeneous equations in n unknowns is a subspace of $\mathbb{R}^m$.

Q.2) Find the null space of $\begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$

Q.3) Let

$$A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$$

- (a) If the column space of A is a subspace of $\mathbb{R}^k$, what is k ?
- (b) If the null space of A is a subspace of $\mathbb{R}^k$, what is k ?

Q.4) Find a 2×2 matrices whose null space is the line $3x - 5y = 0$.

Q.5) Find the complete solution of

$$\begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

Medium

Q.6) Find the complete solution of $A\mathbf{x} = \mathbf{b}$:

$$(a) \begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 \\ 2 & 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & -2 & 3 \\ 2 & 1 & 4 \\ 1 & -7 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ -1 \end{bmatrix}$$

Q.7) Under what conditions on b_1 and b_2 does $A\mathbf{x} = \mathbf{b}$ have a solution?

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 0 & 7 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Find the two vectors in the null space of A and the complete solution to $A\mathbf{x} = \mathbf{b}$.

Q.8) Construct a 2×2 matrix whose null space equals column space. Further justify why no 3×3 matrix will have a null space that equals its column space.

Q.9) Find the null space of

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 & 4 \\ 5 & 5 & 5 & 5 & 5 & 5 \\ 6 & 6 & 6 & 6 & 6 & 6 \end{bmatrix}$$

Q.10) Determine whether the following are subspaces of $\mathbb{R}^3$:

(a) $S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : xy = 0 \right\}$

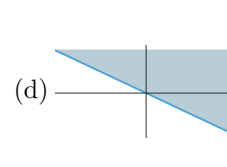
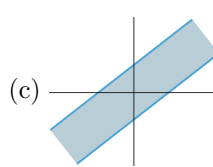
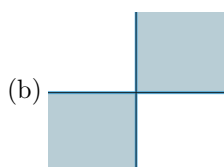
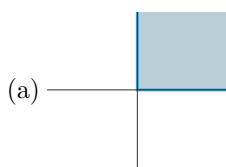
(b) $S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x - y = z \right\}$

(c) $S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x^2 + y^2 = z^2 \right\}$

(d) $S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x^2 + y^2 + z^2 > 0 \right\}$

Hard

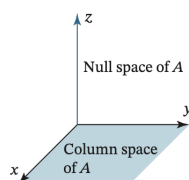
Q.11) Assume that the sets include the bounding lines. In each case, give a specific reason why the set H is not a subspace of $\mathbb{R}^2$.



Q.12) Construct a matrix whose column space contains $(1, 1, 5)$ and $(0, 3, 1)$ and whose null space contains $(1, 1, 2)$.

Q.13) Construct a matrix whose column space contains $(1, 1, 0)$ and $(0, 1, 1)$ and whose null space contains $(1, 0, 1)$ and $(0, 0, 1)$.

Q.14) Find a 3×3 matrix whose null space is the x -axis and whose column space is the yz -plane.



Q.15) Find a 2×3 matrix A such that the complete solution of the system

$$A\mathbf{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{is} \quad \mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \quad \text{where } t \in \mathbb{R}.$$

Challenge

Q.16) Six beams, one molecule

Six laser beams are aimed at a single molecule sitting at the centre. Beam i pushes the molecule with strength $F_i \geq 0$ along a fixed direction $\mathbf{u}_i$. Beams 1 and 4 are horizontal; beams 2, 3, 5, 6 come in at 45° (see figure):

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix}, \mathbf{u}_4 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \mathbf{u}_5 = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}, \mathbf{u}_6 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}.$$

A **pattern** $\mathbf{x} = (F_1, \dots, F_6)$ lists the six strengths. Adding up the six pushes gives the net force $A\mathbf{x}$, where the columns of A are $\mathbf{u}_1, \dots, \mathbf{u}_6$:

$$\underbrace{\begin{bmatrix} 1 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & -1 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}}_A \underbrace{\begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \\ F_6 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}}_{\mathbf{b}}$$

The molecule is **held still** when the net force is zero, that is, when $A\mathbf{x} = \mathbf{0}$. The set of net forces that can be made is $\text{Col}(A)$.

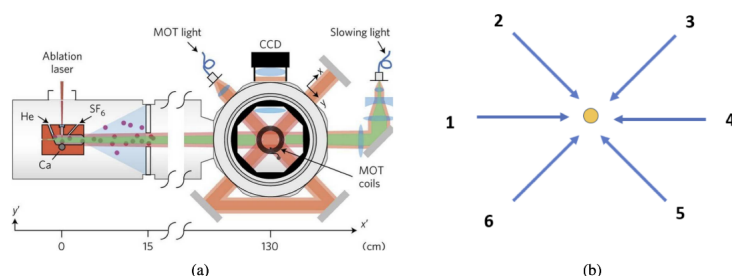


Figure 1: each arrow shows the direction in which that beam pushes.

- Apply Gaussian elimination to A and write down R , the row echelon form. How many pivots are there? What is the rank? Which of $F_1, \dots, F_6$ are pivot variables and which are free?
- Without any calculation, give a one-line reason based only on the *shape* of A why the six directions must be linearly dependent, and why $A\mathbf{x} = \mathbf{0}$ has a solution other than $\mathbf{x} = \mathbf{0}$.
- Patterns that hold the molecule still, found by eye.
 - Check that $\mathbf{x} = (1, 0, 0, 1, 0, 0)$ gives $A\mathbf{x} = \mathbf{0}$: only beams 1 and 4 are on, at equal strength. Write down two more patterns of this same “opposite pair” type.
 - Now set all six strengths equal: $\mathbf{x} = (1, 1, 1, 1, 1, 1)$. Check that $A\mathbf{x} = \mathbf{0}$, and write this pattern as a sum of the three patterns from part (i).
 - Beam 4 stops working, so $F_4 = 0$. Show the molecule can still be held still: find a pattern with $F_4 = 0$ and all five other strengths strictly positive.
(Hint: beams 3 and 5 both push in the $-x$ direction. Use them against beam 1, then add an opposite pair to switch on the beams that are still off.)
- In one sentence, say what $\mathcal{N}(A)$ means here physically.
- The molecule now has to be pushed sideways with net force $\mathbf{b} = (1, 0)$.

- (i) Write down the obvious one-beam recipe $\mathbf{x}_p$ with $A\mathbf{x}_p = \mathbf{b}$, and check it.
- (ii) Using the free variables from part (a), write the complete solution of $A\mathbf{x} = (1, 0)$ in the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$, where $A\mathbf{x}_n = \mathbf{0}$. If the strengths may be any real numbers, how many recipes give this force?
- (iii) Real lasers can only push, so in fact $F_i \geq 0$ for every i . Show that there are still infinitely many usable recipes by adding a multiple of $(1, 1, 1, 1, 1, 1)$ to $\mathbf{x}_p$. What does this say about turning all six beams up together?
- (iv) In one sentence: what is the difference in role between $\mathbf{x}_p$ and $\mathbf{x}_n$?
- (f) *True or false? Give a reason.*
 - (i) Adding a seventh beam would make $\text{Col}(A)$ bigger.
 - (ii) If two patterns each hold the molecule still, so does their sum.
- (g) **Challenge.** Keeping the restriction $F_i \geq 0$, can every net force $\mathbf{b} \in \mathbb{R}^2$ still be made? Argue yes, and give a rule for building a recipe with no negative strengths for any $\mathbf{b}$.

Linear Algebra

LAB Sheet - Lecture 8

Name: _____

ID/Batch: _____

Easy

Q.1) State True/False. Justify your answer.

- (a) If $m = n$ then the row space of A equals the column space of A .
- (b) The matrices A and $-A$ share the same four subspaces.
- (c) If A and B share the same four subspaces then A is a multiple of B .
- (d) A and A^T have the same number of pivots.
- (e) If the row space equals the column space then $A^T = A$.
- (f) If B is any echelon form of A , then the pivot columns of B form a basis for the column space of A .
- (g) The dimensions of the row space and the column space of an $m \times n$ matrix A are equal, even if A is not square.

Q.2) Find the basis of column space and null space of the following matrix. Further determine the rank and nullity.

a) $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 2 & 5 & 8 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 5 & 1 & 4 \\ 3 & 7 & 0 & 7 \\ 4 & 9 & -1 & 10 \end{bmatrix}$

Q.3) Answer each part and explain your reasoning.

- (a) If A is a 7×5 matrix, what is the largest possible rank of A ? If A is a 5×7 matrix, what is the largest possible rank of A ?
- (b) If A is a 4×3 matrix, what is the maximum possible dimension of the row space of A ? If A is a 3×4 matrix, what is the maximum possible dimension of the row space of A ?
- (c) If the null space of a 5×6 matrix A is 4-dimensional, what is the dimension of the row space of A ?
- (d) If the null space of a 7×6 matrix A is 5-dimensional, what is the dimension of the column space of A ?

Q.4) In each part, use the information in the table to find the dimension of the row space of A , column space of A , null space of A , and left null space of A .

	(a)	(b)	(c)	(d)	(e)	(f)
Size of A	3×3	3×3	3×3	5×9	9×5	4×4
$\text{rank}(A)$	3	2	1	2	2	0

Q.5) Find the complete solution $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ of $A\mathbf{x} = \mathbf{b}$ for

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 5 \\ 5 \end{bmatrix}.$$

State the rank, the nullity, and the number of free variables, and check that no. of columns = rank(A) + nullity(A).

Medium

Q.6) Matrix B is the row echelon form of matrix A . Without calculations, list rank A and $\dim(N(A))$. Then find bases for $C(A)$, $C(A^T)$, and $N(A)$.

$$A = \begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ -2 & 3 & -3 & -3 & -4 \\ 4 & -6 & 9 & 5 & 9 \\ -2 & 3 & 3 & -4 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ 0 & 0 & 3 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Q.7) Given $A = LU$, without computing A , find bases for the four fundamental subspaces:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 9 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Q.8) Find the dimension and construct a basis for the four subspaces associated with each of the matrices

$$A = \begin{bmatrix} 0 & 1 & 4 & 0 \\ 0 & 2 & 8 & 0 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Which of the four subspaces are the same for A and U , and which are different? Explain why.

Q.9) Under what conditions on b_1, b_2, b_3 is the following system solvable? Augment $\mathbf{b}$ as a fourth column and perform elimination. Find all solutions when that condition holds.

$$x + 2y - 2z = b_1$$

$$2x + 5y - 4z = b_2$$

$$4x + 9y - 8z = b_3$$

Then find a vector $\mathbf{y}$ spanning $N(A^T)$ and verify that your solvability condition is exactly the statement $\mathbf{y}^T \mathbf{b} = 0$.

Q.10) If V is the subspace spanned by $(1, 1, 1)$ and $(2, 1, 0)$, find a matrix A that has V as its row space. Find a matrix B that has V as its null space.

Q.11) Answer each part, with an explanation.

(a) Suppose a 4×7 matrix A has four pivot columns. Is $C(A) = \mathbb{R}^4$? Is $N(A) = \mathbb{R}^3$?

(b) Suppose a 5×6 matrix A has four pivot columns. What is $\dim(N(A))$? Is $C(A) = \mathbb{R}^4$? Why or why not?

Q.12) Are there values of r and s for which

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & r-2 & 2 \\ 0 & s-1 & r+2 \\ 0 & 0 & 3 \end{bmatrix}$$

has rank 1? Rank 2? If so, find those values.

Hard

Q.13) If the product of two matrices is the zero matrix, $AB = 0$, show that the column space of B is contained in the null space of A . (Also the row space of A is contained in the left null space of B .) Why can't A and B be 3×3 matrices of rank 2?

Q.14) Construct a matrix with the required property, or explain why this is impossible.

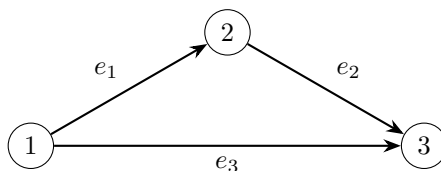
- (a) Column space contains $(1, 1, 0)$ and $(0, 0, 1)$; row space contains $(1, 2)$ and $(2, 5)$.
- (b) Column space has basis $(1, 1, 3)$; null space has basis $(3, 1, 1)$.
- (c) Dimension of null space = 1 + dimension of left null space.
- (d) Left null space contains $(1, 0, 0, 0)$; row space contains $(0, 0, 0, 1)$.
- (e) Row space = column space, but null space $\neq$ left null space.

Q.15) Suppose A is an $m \times n$ matrix of rank r . Under what conditions on those numbers does

- (a) A have a two-sided inverse, $AA^{-1} = A^{-1}A = I$?
- (b) $A\mathbf{x} = \mathbf{b}$ have infinitely many solutions for every $\mathbf{b}$?

Q.16) A is an $m \times n$ matrix of rank r . Suppose there are right sides $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has no solution. What are all the inequalities ($<$ or $\leq$) that must be true between m , n , and r ? How do you know that $A^T\mathbf{y} = \mathbf{0}$ has solutions other than $\mathbf{y} = \mathbf{0}$?

Q.17) The graph below has 3 nodes and 3 edges. Write down its 3×3 incidence matrix A , using *one row per edge*: put -1 in the column of the node the edge leaves and $+1$ in the column of the node it enters. (Note that this is the transpose of the convention used in the lecture sheet, where one column stood for one edge.)



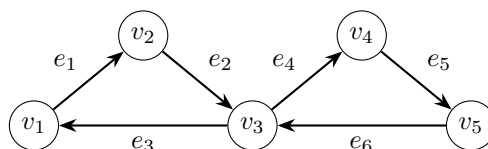
- (a) Find a solution to $A\mathbf{x} = \mathbf{0}$ and describe all other vectors in the null space of A . What do the entries of $\mathbf{x}$ measure here?
- (b) Find a solution to $A^T\mathbf{y} = \mathbf{0}$ and describe all other vectors in the null space of A^T . What do the entries of $\mathbf{y}$ measure here?
- (c) Show directly from the columns of A that every $\mathbf{b}$ in the column space satisfies $b_1 + b_2 - b_3 = 0$. Then derive the same condition from the three equations of $A\mathbf{x} = \mathbf{b}$.

Challenge

Q.18) Six pipes, one water grid

A campus water grid has 5 junctions $v_1, \dots, v_5$ joined by 6 pipes. Each pipe has been given a reference direction (the arrows below); a flow f_i is positive when water runs along the arrow and negative when it runs against it.

$$e_1 : v_1 \rightarrow v_2, \quad e_2 : v_2 \rightarrow v_3, \quad e_3 : v_3 \rightarrow v_1, \quad e_4 : v_3 \rightarrow v_4, \quad e_5 : v_4 \rightarrow v_5, \quad e_6 : v_5 \rightarrow v_3.$$



Store the grid in the incidence matrix B with *one row per junction, one column per pipe*, as in the lecture sheet: -1 where a pipe leaves a junction, $+1$ where it enters. Then $B\mathbf{f}$ is the vector of *net inflows* at the five junctions: positive means water accumulates there (a consumer draws it off), negative means water is supplied there.

- Write down B . What is its size? Which space is the input home and which is the output home, and what does a vector in each one physically represent?
- Reduce B and find $\text{rank}(B)$. Pick out 4 pipes that connect all five junctions without closing a loop (a *spanning tree*), and use it to explain why a connected graph on n nodes has $\text{rank}(B) = n - 1$.
- Find a basis for $N(B)$ and mark each basis vector on the diagram. What does each one do physically? Verify that

$$\text{nullity}(B) = (\text{number of pipes}) - (\text{number of junctions}) + 1,$$

and say why this counts the independent loops in the grid.

- Find a basis for $N(B^T)$. Since B^T turns junction pressures into pipe pressure-drops, state in one sentence what this space says about pressure.
- Which demand vectors $\mathbf{d} \in \mathbb{R}^5$ can the grid actually meet, i.e. for which $\mathbf{d}$ is $B\mathbf{f} = \mathbf{d}$ solvable? Use your basis from (d) to compress the answer into a single scalar equation, and confirm that

$$\mathbf{d} = (10, 0, -4, 0, -6)$$

passes the test. Find the complete solution $\mathbf{f} = \mathbf{f}_p + \mathbf{f}_n$ for this $\mathbf{d}$ by setting $f_3 = f_6 = 0$. How many flow patterns meet this demand? Interpret the negative entries of $\mathbf{f}_p$.

- A vector $\mathbf{w} \in \mathbb{R}^6$ of pressure drops comes from junction pressures $\mathbf{p}$ exactly when $\mathbf{w} = B^T\mathbf{p}$ is solvable, i.e. when $\mathbf{w} \in C(B^T)$ — the row space. Using your loop basis from (c), show that this happens precisely when the entries of $\mathbf{w}$ sum to zero around each loop. Which physical law is this?
- Pipe e_3 is shut for maintenance, forcing $f_3 = 0$. Delete column e_3 from B to get B' .
 - Give $\text{rank}(B')$, $\text{nullity}(B')$ and $\dim N(B'^T)$.
 - Can the demand $\mathbf{d}$ of part (e) still be met? If so, how many flow patterns meet it now?
 - State which of the four subspaces changed and which did not, and explain each answer in one line.